

KhelQuest: AI-Powered Platform for Democratizing Sports

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Abstract—Many talent identification methods utilized in traditional sports are oftentimes hindered by high costs, manual evaluation processes, access limitations (especially in rural areas), and largely unavailable tools to facilitate performance assessment. This paper introduces an AI-Powered Platform for Democratizing Sports: a mobile-based system that allows for scalable, real-time, and objective assessments of an athlete's performance. The system employs computer vision and machine learning techniques via pose estimation (which uses the MediaPipe SDK) to extract the body keypoints from video input, and to accurately analyze movements. An on-device TensorFlow Lite model enables real-time processing with mobile devices, providing efficient counting of repetitions, analysis of posture, and performing evaluation; along with eliminating the need for continuous internet access throughout the assessment process. Key performance metrics (speed, endurance, technique, consistency) are computed using rule-based and data-driven techniques, in addition to detecting anomalies for the purpose of identifying any cheating. In addition to performance assessment, the platform provides several other features to support athlete development, such as generating AI-driven training plans, connecting with an AI coach, making contacts with other athletes as well as coaches and sports authorities; providing access to playgrounds; etc. The backend of the platform utilizes a combination of Node.js, FastAPI, and cloud-based databases to support secure data management and to generate rankings at various levels. Overall, the proposed system is an inclusive, low-cost, scalable solution that bridges the gap between grassroots athletes and professional opportunities, thereby supporting fair and accessible access to identify athletes with sports talent.

Index Terms—Artificial Intelligence, Computer Vision, Sports Analytics, Pose Estimation, MediaPipe, TensorFlow Lite, Athlete Performance Analysis, Talent Identification, Smart Sports Systems

I. INTRODUCTION

Recently, advancements in Artificial Intelligence (AI) and Machine Learning (ML) are now being used in many different industries, such as Healthcare, Education, and Sports Analytics. AI-based systems are being used more and more to analyze human movement and improve performance through

data-driven decisions [1]. Sports traditionally rely on manual observations, expensive biomechanical equipment, central training locations, which limits accessibility for many athletes (especially in rural areas/poorer communities) [2].

Conventional sports evaluation methods have complex specialized infrastructure (motion capture systems, force plates, wearable devices); can be expensive; and therefore, costly and challenging to scale [3]. Furthermore, using human assessments can add subjectivity and bias into the equation, causing inconsistencies and non-transparency [4]. Thus, there is a clear need for an automated, scalable, unbiased performance evaluation/talent-identification system for sports.

The use of Computer Vision and Deep Learning Techniques for athletic performance assessment by using video footage is being explored in recent research. Decentralized models such as OpenPose and MediaPipe have accurately identified key human body points/persons while simultaneously tracking their movement in real-time [5][6]. Alternative approaches to measuring motion and analyzing movement through non-physical means do not require sensors or markers that are placed on an athlete's body and therefore makes it a viable and cost-effective option for widespread use. Studies have shown that both smartphone-based systems and laboratory-based systems have similar accuracy levels when using optimal machine learning models together.

Recent advancements in machine learning have also expanded the horizons of on-device ML frameworks with the introduction of on-device ML systems like TensorFlow Lite, which allow for instantaneous data processing (like fitness counting, correcting form, recognizing activity, etc.) directly on a mobile device without having to rely on the Internet for a connection. Other ML technologies like identifying bad movements (anomaly detection) and utilizing classification models (classification learning) have been developed and are being used to evaluate athletes in a fair manner. So to address these challenges, this paper proposes an AI-Powered Platform

for Democratizing Sports, which integrates computer vision, mobile computing, and cloud-based analytics into a unified system.

II. LITERATURE REVIEW

Until recently, there has not been a way to automate sports motion analysis due to the lack of either the technology or the financial resources to build such a system. Now, there is an increasing reliance on video-based approaches to analyzing athletic performance, which have provided many advantages over traditional manual observation and costly equipment, including the ability to scale, cost savings [11].

Automated motion analysis using vision-based technologies does not require the physical use of sensors and allows for evaluating athletic performance through motion capture. Many studies have shown that various exercises can be evaluated, including squats, push-ups and jumps by using camera-based systems, and can achieve a reasonable level of accuracy, enabling large scale implementation of evaluation systems [12].

Pose estimation techniques have greatly enhanced the analysis of an athlete's performance. For example, real-time detection of human body keypoints has been achieved using model-based systems such as OpenPose and MediaPipe. This allows for the detailed tracking of joint movement and positioning, which serves as the basis for automated applications such as counting repetitions or correcting the athlete's posture or activity recognition [13].

There is a substantial body of work on developing algorithms that use machine learning for activity classification and performance evaluation metrics. The use of these algorithms has permitted the automation of evaluation processes, reduced humans' bias and provided greater consistency in evaluation [14].

Recent advances in mobile-based artificial intelligence systems have made these types of technologies and evaluations accessible to a wider audience. One of the key frameworks enabling machine learning to support athletes directly on smartphones, enabling instant processing with minimal reliance on internet connectivity [15], is TensorFlow Lite; which can greatly aid applications associated with use in rural locations with few additional resources.

Most available systems that provide performance analyses capture little more than performance statistics and do not provide features such as personalized training programs, athlete networking, or coach/sports organisation connections. Also, most existing performance analysis systems have been created for controlled environments, thus do not provide for scalability or inclusion [16].

III. OBJECTIVES

The main goal of this solution is to make it easy for athletes to find good talent to train and improve. Some ways that we will do this are:

A. Accessible Talent Identification

The goal is to create a system that allows athletes to be evaluated at home while using their smartphones or similar devices to take standardized fitness tests. This technology will eliminate the need for expensive facilities to help identify talent. It also allows talent identification to involve individuals from all economic levels and backgrounds, not just those who can afford it[1][15].

B. AI-Based Performance Evaluation

You can use new technologies, like the ones used by computers to identify people's locations and evaluate their movements, to measure performance or technical skills (for example, with computer vision and machine learning techniques). You will use these technologies to automatically evaluate how well an athlete has performed on attributes related to speed, endurance, technique and consistency (measured as a function of time)[5][13].

C. Real-Time Feedback and Improvement

The use of analysis using AI technology will provide immediate feedback to help athletes make corrections so that they will constantly know how well they are performing and the ways in which they can improve without having someone supervising them constantly [14].

D. Personalized Training and AI Coaching

A system will generate customized training plans, which will include an AI coach that will help each athlete based on their performance and trends over time [15].

E. Fair and Transparent Evaluation

Performance evaluations will be fair and impartial by decreasing human involvement and using techniques to detect anomalies to identify any performance or behavior that was either false or changed. [9]

F. Athlete Networking and Collaboration

By providing a place for athletes to connect, track their progress, and learn from each other and from their mentors, the program will build a collaborative environment for sports [10].

G. Integration with Coaches and Sports Authorities

There will be a way for athletes, coaches and sports authorities to connect so they may receive training and mentorship and develop an official selection process [2].

H. Scalable and Low-Cost Deployment

The system will be designed to operate properly on low-end devices with minimal use of the Internet for connectivity so that it can also operate in rural and other areas where there are limited resources [8][15].

TABLE I
COMPARATIVE ANALYSIS OF EXISTING WORKS AND PROPOSED SYSTEM

Ref.	Methodology	Technology	Deployment	Limitation	KhelQuest Advantage
[1]	Survey-based review of AI in sports	General ML and analytics models	Conceptual research studies	No real-world system implementation	Provides practical mobile-based system
[2]	Video-based ML performance evaluation	Computer vision + ML classification	Controlled indoor setups	Limited scalability discussion	Works with smartphone videos in field
[3]	Deep learning for talent detection	CNN/DNN trained on labeled datasets	Structured dataset environments	High dependency dataset	Operates on real-world recordings
[4]	Lightweight model optimization for edge	TinyML and quantized networks	Low-resource embedded devices	Performance trade-offs in scaling	Combines lightweight AI with cloud support
[5]	Cloud-based sports analytics dashboards	Web systems + database analytics	Professional sports institutions	Not designed for grassroots usage	Targets rural and decentralized athletes
[6]	Secure data management framework	Encryption and cloud authentication	Mobile health applications	Not sports-specific	Integrates security into sports workflow
[7]	Fairness and bias analysis in AI	Bias detection and transparency models	Theoretical research focus	Limited practical integration	Implements fairness in scoring system
[8]	Pose-based activity recognition	Skeletal tracking + vision models	Specific exercise tracking	Only classification, no ranking	Adds scoring and benchmarking
[9]	Vision-based activity recognition	Deep CNN pose modeling	Controlled validation datasets	Sensitive to environment changes	Designed for real-world field usage
[14]	Video tampering detection framework	Deep learning + multimedia forensics	Digital content verification	High computational complexity	Lightweight anti-cheating validation
Proposed	End-to-end AI sports assessment pipeline	Pose + ML + Cloud + Mobile	Grassroots, decentralized environments	Depends on video quality conditions	Integrated, scalable, inclusive ecosystem

I. Data Security and Verified Profiles

Athlete's information will be kept secure, and a verified digital profile will be developed based on the athlete's performance, so it can be utilized to track the athlete and recognize their achievements [6].

IV. SYSTEM DESIGN

The architecture of the KhelQuest Platform (the proposed system) is illustrated in Figure 1. The figure indicates the original functionality as well as the flow of information between user levels. A 3-tier architecture consists of the client tier, middle tier, and data tier to help maintain modularity, scalability, and efficient processing.

The system model for the proposed KhelQuest Platform is illustrated in Figure 2. The system model defines how users will interact with the KhelQuest Platform and what the platform will look like after being implemented. The use case diagram shows user interactions and the class diagram defines the relationships between the class entities in the KhelQuest Platform.

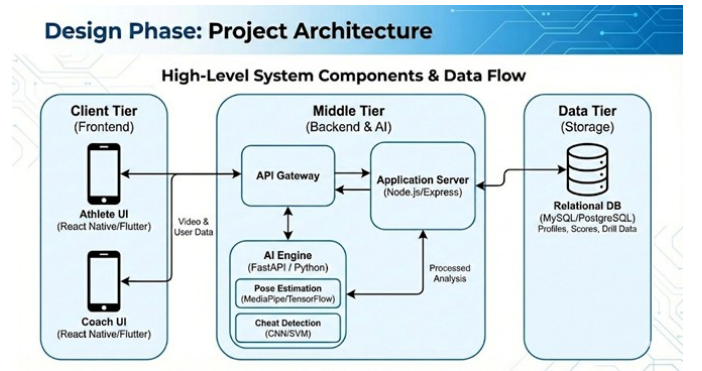


Fig. 1. System Design

V. METHODOLOGY

The approach taken in this work will facilitate. objective and available sports talent measurement through artificial intelligence and video analysis using smart phones. The general

System Modeling: Structure & Interactions

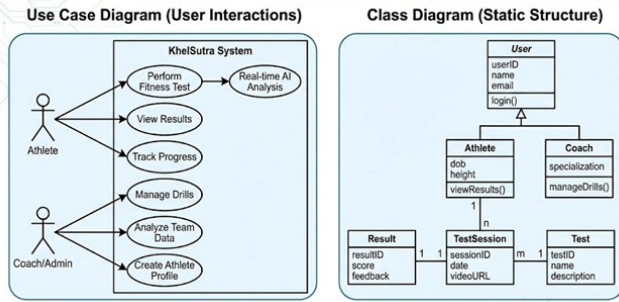


Fig. 2. System Model

structure is aimed at reducing the use of manual appraisal and expensive infrastructure and assurance of equity, scalability, and performance analysis that can be measured [1]. The is an artificial intelligence method that combines machine learning, computer vision and. systematized information organization into a single analysis process. This begins with the sportsmen executing predetermined. vertical jumps or push-ups, which are fitness exercises. are captured on one of the mobile devices.

The main input into the system is these recorded videos. Basic metadata, Each is associated with including user role and test type. session in order to have structured assessment. This smartphone-based data acquisition provides availability and engagement in all directions. diverse environments. After the acquisition of data, computer vision methods are. used to obtain pose-based representation of body movements. Pose estimation models discover and track important skeletal. this is made possible by joint coordinates between successive video frames. precise posture, motion path and serialization modeling. patterns [5]. This reduces the dependency due to this skeletal abstraction. on background conditions and enhances robustness against variations in lighting or camera angle. Based on the pose sequences obtained, there are pertinent motions. features are computed. Such characteristics might incorporate joint angle. variations, time-consistency, pattern of displacement, and repetition counts. Analysis is done on the extracted features. using machine learning models trained to evaluate movement. Based on the pose sequences obtained, there are pertinent motions. features are computed. Such characteristics might incorporate joint angle. variations, time-consistency, pattern of displacement, and repetition counts.

Analysis is done on the extracted features.using machine learning models trained to evaluate movement quality and performance measures. The evaluation process em- stresses on quantifiable and objective standards instead of subjective ones. Active human observation and, thus, minimizing the evaluator bias. refining the assessment consistency [8].

Progress of an athlete can be monitored using historical records. preservation through time and support data-driven decision-making via. coaches and administrators [11], [12]. The overall methodology focuses on ethical AI deployment, inclusivity and scalability, which makes a solid basis of the

system implementation This is station as explained in the following section.

VI. IMPLEMENTATION

The AI-Driven Sports Democratization Platform (KhelQuest) will be a mobile-based platform that comprises the following three major systems: a front-end user interface, a back-end processing system, and an artificial intelligent analysis module. This platform is designed to provide users with uninterrupted usability, rapid process of capturing, and secure storage of data for purposes of evaluating and quantifying an individual's sports performance using smartphones, tablets, and other consumer-level devices ([1], [5]).

A. User Interaction and Registration Process

The user interaction and registration process is conducted according to a set procedure, which is shown in Exhibit 3. Each user registers for an account through the mobile app by entering the required information along with their type of account (athlete/coach).

Once the user's information is received, the back-end system performs verification of the information provided and validates the role of the user. This creates an authenticated account for that individual, appropriately creates the account's profile, and controls the level of access to the features of the platform based upon the user's account type, which includes performance tracking, coaching, and connecting to other athletes and coaches ([4]).

B. Test Procedures and Flow of Information

The data flow between each user and the back-end systems is executed in a step-by-step manner, as illustrated in Exhibit 4. Specifically, when an athlete would like to perform a test of their strength and conditioning (i.e., amount of push-ups, squats, or vertical jumps), the mobile application will capture video from the athlete using the device camera.

The captured video frames will be uploaded to the AI Analysis module and analyzed using pose analysis to capture key body positions and to track the athlete's movements (i.e., to evaluate the number of repetitions); as well as, collect data related to the proper technique of performing the individual exercise. The final results are displayed on the athlete's dashboard in an intuitive format [8], [11].

C. Backend Processing and Data Management

The backend of the system handles all API requests, authentication, session management and secure storage of athlete data.

The system effectively stores and processes a structured database containing each athlete's profile, performance record, ranking and achievements. The modular nature of the backend design allows for scalability in terms of concurrent users, as well as accommodate future upgrades such as advanced analytics or integration with sports authorities [5], [12].

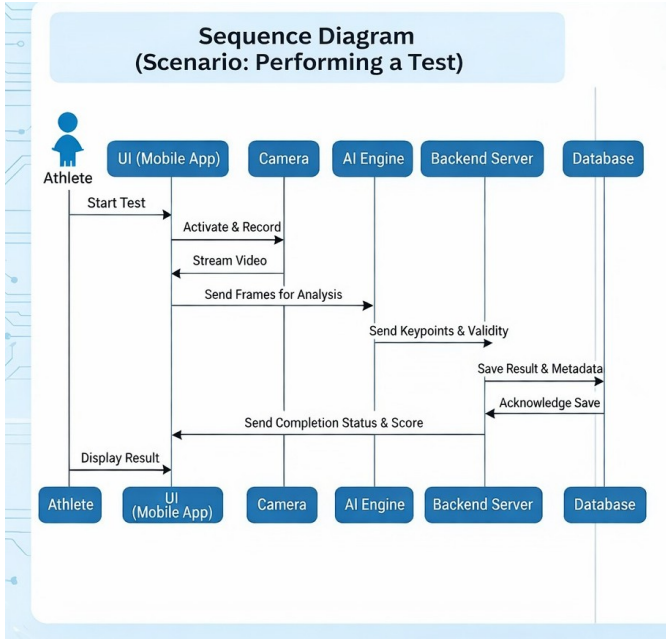


Fig. 3. Implementation

D. AI Model Optimization and Deployment

The AI model deployment strategy shown in Fig. 6 will ensure efficient execution on mobile devices by utilizing light-weight, TensorFlow Lite optimized models. Additionally, model quantization and architecture optimization reduce computational complexity while maintaining model accuracy [3],[9].

The hybrid processing approach will be employed across the system, including:

On-Device Processing: Real-time pose estimation, rep counting, and immediate feedback **Cloud Processing:** Advanced analytics, ranking generation, and long-term tracking Hybrid processing minimizes latency, increases responsiveness and enhances scalability [8], [11].

E. System Usability and Deployment Considerations

Usability in real world environments, particularly rural and resource constrained areas is a significant focus for the deployment strategy.

Reliability in low network conditions will ensure large scale adoption. The optimized AI pipeline and mobile friendly architecture make the system practical, scalable and sustainable for decentralized sports talent assessment.

VII. RESULTS AND DISCUSSIONS

A. Results

The KhelQuest platform has undergone testing through various fitness assessment tasks completed by athletes of all ages and from various geographical locations. The KhelQuest platform was able to accurately compare performance between participants (e.g., recording and analyzing video to generate an

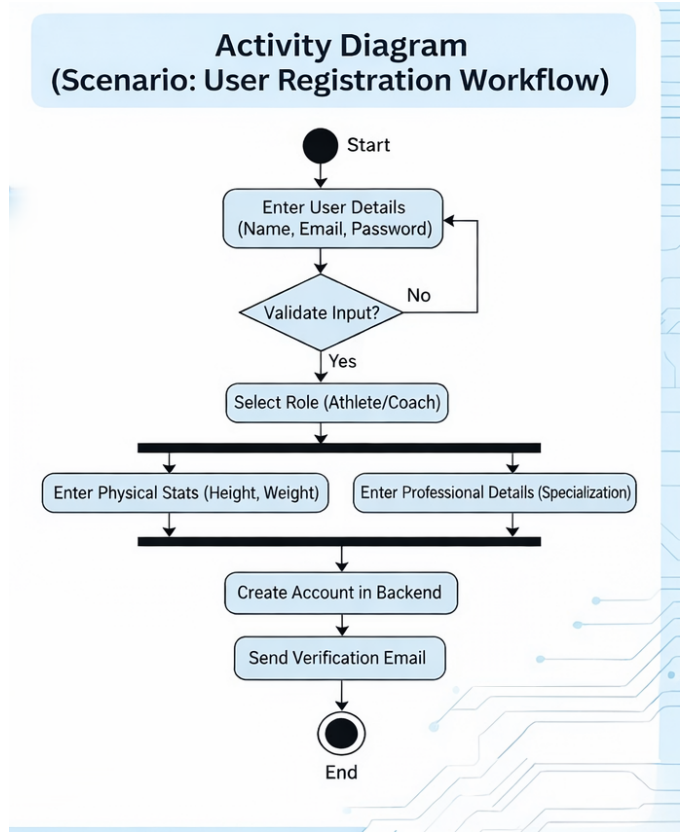


Fig. 4. Implementation

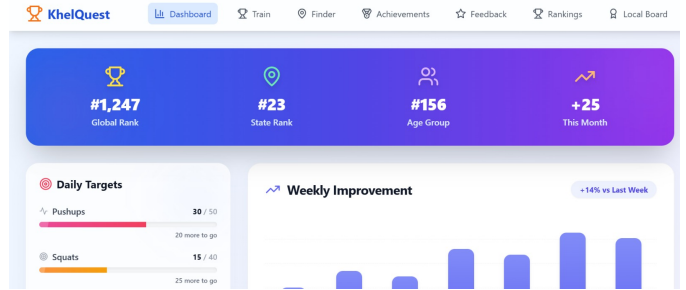


Fig. 5. Dashboard

accurate score, ranking and providing performance feedback without the use of external sensors, or manual input).

Esports has multiple levels of ranking (district, state and worldwide) as shown in Figure 4. This enables an athlete to assess performance in relation to all peers, taking into account their age group as well as their skill level when comparing their ranking ranking scores (i.e., the scores would be dynamically updated based on test scores) and also provides progress indicators for monitoring improvements or declines in performance over time.

The evaluation data supports the ability of the KhelQuest platform to provide multi-level comparisons of performance. There was also a consistent differential in the performance rating scores of the athlete, as indicated by the consistency

in measurement of movement quality and accuracy with how each athlete completed each exercise. In addition, all response time to perform analysis was stable, and the KhelQuest platform was able to store each athlete's performance based on how long they have performed, enabling a longitudinal comparison of performance.

In summary, the findings of the evaluation confirm that the use of smartphone technology for conducting AI-based sports performance evaluations are available and feasible in resource-constraint / decentralized environments[1], [5].

B. Discussions

The study explores how effective the computer vision alongside machine learning technology are when assessing sports performance in a consistent and impartial manner.[12].

Athletes are motivated to improve their performance by understanding their relative performance to their competitors by using the new system, which is made possible through the use of comparative results and performance benchmarks. The ability to determine how athletes are progressing and depending upon their rankings, has been shown to improve the level of engagement for those individuals and promote consistent training and practice levels[4].

The data visualization capabilities of the dashboards produced by the computer vision and machine learning based assessments will assist coaches and sport administrators in making decisions based upon objective data. The storage and tracking of athletes' performance data by the system enables coaches and sport administrators to continue monitoring and evaluating the athletes long-term which is critical to the long-term success of athlete development programs[8].

VIII. CONCLUSION AND FUTURE SCOPE

A. Conclusion

The KhelQuest platform is designed to democratise the identification of sport talent using artificial intelligence and smartphone video analysis. The platform uses computer vision and machine learning techniques to automate and objectively evaluate sporting performance, eliminating the need for special infrastructure and costly materials [2][3].

The KhelQuest platform reduces barriers of entry to sports and exposes more youth to participation in sport, especially those athletes from rural and underrepresented areas. The automated and unbiased scoring system allows all users to be equally evaluated and provides all users with a reliable performance evaluation [4][5].

B. Future Scope

1) *Advanced Predictive Modeling and Talent Projection:* Future system enhancements will be developed to include Predictive Models to conduct longitudinal trend analyses of athlete performance. These models will support early identification of athlete talent and allow for structured developments plans for athletes based on a historical record of performer and development trends [3].

2) *Integration of Wearables and Multimodal Sensors:*

Integration of multiple wearable sensor types (i.e. accelerometers, heart-rate monitors) to provide additional context to athlete performance data will provide increased accuracy and contextualization of data captured from athlete performance in sports analytics systems [4], [5].

3) *Enhanced Anti-Cheating Mechanisms:* Additional research can include exploration of additional methods to detect anomalies and verify the validity of recorded performance data with an emphasis on advanced methods for conducting these activities. Strengthening the reliability, integrity and trustworthiness of a system will create greater value through competitive evaluations.

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